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Supersonic Single Wedge Airfoil

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Introduction

Consider a high performance sounding rocket. No, not what's called "High Performance" by the amateur rocket community, but, what's called high performance by the professional rocket community. In the latter case, well developed supersonic flow is the norm, especially around burnout. Such rockets commonly present a stability challenge to their designers: During burning the Center of Gravity moves forward as propellant is consumed. But, the Center of Pressure moves even further forward because the forebody normal force increases with increasing supersonic Mach number while that on the fins decreases with increasing Mach number[1].

This note describes one of the design tricks of the aerodynamicist's trade often used to mitigate this stability problem. It is simple: replace the ordinary flat plate airfoil with a single wedge airfoil (pointy edge forward). There is some limited data, see ref's (2) and (3), showing no significant loss of subsonic lift because the Kutta condition still keeps the circulation up. There will be some base drag at all Mach numbers and an increase in supersonic wave drag that will make this airfoil unsuitable for aircraft applications. However this is only a minor consideration for rockets. Finally, it also provides the designer with interior room for structure, often a major plus. And, a flat fin surface facilitates cant angle adjustments.

Analysis

The usual point of departure is the equations for an oblique shock wave. See eq. (152) of ref. (1). This relates the free stream Mach number M_1, to the pressure coefficient Cp on a wedge of angle δ. Here, if the leading edge is swept, δ is defined in a plane normal to the leading edge. Similarly, the Mach number M_1 here is taken normal to the leading edge. The leading term in this Cp expansion is just that derived by Ackeret. Busemann suggested that the series be truncated after the second term resulting in

$$Cp = \frac{2\delta}{\beta^{1/2}} + \frac{\left[(\gamma+1)M_1^4 - 4\beta\right]\delta^2}{2\beta^2} \qquad (1)$$

where $\beta = M_1^2 - 1$, and

[1] To a lesser degree the same thing happens to vehicles stabilized with a flare in lieu of fins

γ = Ratio of specific heats for air = 1.4.

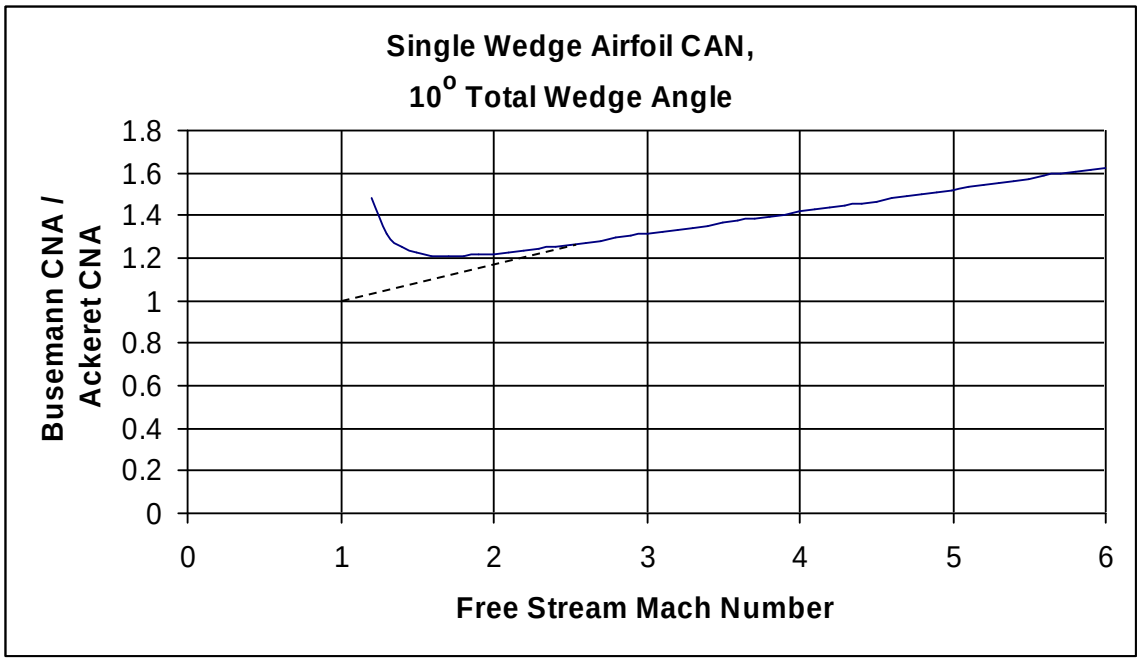
The rest is easy. Suppose δ were increased by a small angle of attack α on the compression (lower) side of a single wedge airfoil, and decreased by the same small angle of attack on the expansion (upper) side. The difference in Cp between the two sides is the airfoils normal force coefficient slope $C_{N\alpha}$:

$$C_{N\alpha} = \frac{4}{\beta^{1/2}} + \frac{2\left[(\gamma+1)M_1^4 - 4\beta\right]\delta}{\beta^2} \tag{2}$$

In practice the ratio of the Busemann (quadratic) result to the linear Ackeret provides a multiplier that can be applied to any single wedge fin with a supersonic leading edge.

$$\frac{Busemann}{Ac\ker et} = 1 + \frac{\left[(\gamma+1)M_1^4 - 4\beta\right]\delta}{2\beta^{3/2}} \tag{3}$$

This can be clarified with a numerical example



The sharp upturn at Mach numbers less than about 2 is an artifact of the analysis. It is well known that both linear and second order theories have a singularity at M_1. To address this, use eq. (3) only for M_1s greater than 2.5. When $1 \le M_1 < 2.5$, fit a straight line between the $M_1 = 2.5$ calculated point and unity when $M_1 = 1$. This is depicted as a dashed line in the figure. Note that the benefit of a single wedge airfoil tends to be greatest at higher Mach numbers.

The additional wave drag C_{DW} of a wedge airfoil relative to a flat plate can be found from eq. (1):

$$C_{DW} = \frac{4\delta^2}{\beta^{1/2}} + \frac{\left|(\gamma+1)M_1^4 - 4\beta\right|\delta}{\beta^2} \qquad (4)$$

Note that this result does not include the base drag increment. The implication is that there is a design trade to be made. A larger δ will improve the tail normal force and stability, but at the cost of increased wave drag.

Finally, consider what happens to a single wedge airfoil at subsonic speeds. It is natural to fear what might happen when a clearly defined Kutta condition at the trailing edge is lost. Surprisingly, the Kutta condition is very robust. References (2) and (3) show that there is no significant loss in normal force when the trailing edge is terminated bluntly. There will be, of course, a base drag increment.

References

1. Ames Research Staff, "Equations, Tables, and Charts for Compressible Flow", NACA Report 1135, 1953
2. Smith, H. A. and Schaefer, R. F., "Aerodynamic Characteristics at Reynolds Numbers of 3.0×10^6 and 6.0×10^6 of Three Airfoil Sections Formed by Cutting Off Various Amounts from the Rear Portion of the NACA 0012 Airfoil Section", NACA TN 2074, April 1950
3. Summers, J. L. and Page, W. A., "Lift and Moment Characteristics at Subsonic Mach Numbers of Four 10-Percent Thick Airfoil Sections of Varying Trailing-Edge Thickness", NACA RM A50J09, 20 December 1950